

RESEARCH ARTICLE

Ultrawide Charge-Trap Memory Window and Photoinduced Synaptic Behavior in *p*-Channel Amorphous Oxide Semiconductors

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ABSTRACT

The decades-long absence of high-performance *p*-type amorphous oxide semiconductors (AOSs) remains a critical bottleneck in complementary circuit development and has severely limited exploration of charge trapping phenomena essential for neuromorphic computing. Here, we achieve a transformative breakthrough by demonstrating a *p*-channel amorphous oxide semiconductor through ultraviolet-ozone oxidation of crystalline tellurium to amorphous tellurium trioxide (a-TeO₃). This revolutionary material integrates transistor, nonvolatile memory, photodetection, and synaptic functions in a single device, achieving an unprecedented level of functional integration. The a-TeO₃ channel exhibits an unprecedented ultrawide memory window exceeding 58 V under ambient conditions, driven by oxygen vacancy-adsorbate interactions that enable robust multilevel switching. Ultraviolet illumination induces persistent photocurrent through carrier trapping/detrapping, enabling light-programmable synaptic plasticity and associative learning. Paired with monolayer molybdenum disulfide *n*-mode charge-trap memory (CTM), antagonistic charge-trap dynamics realize autonomous heating/cooling control and precise homeostasis. Hardware-constrained networks built on these complementary synaptic transistors achieve MNIST accuracy comparable to ideal digital systems. This light-reconfigurable *p-n* platform overcomes a critical materials barrier, unlocking scalable, energy-efficient neuromorphic architectures for edge artificial intelligence.

1 | Introduction

High-performance *p*-channel oxide semiconductors remain a critical challenge for advancing complementary logic, high-density memory, and neuromorphic computing systems. The *n*-type amorphous oxide semiconductors (AOSs), such as amor-

phous In-Ga-Zn-O (a-IGZO), have already achieved industrial adoption owing to their excellent low off-current and long-term stability [1–6]. In contrast, the development of their *p*-type counterparts has lagged significantly behind. The key obstacle lies in the intrinsically localized O 2p valence bands, which severely limit hole transport [7, 8]. As a result, the lack of efficient *p*-type

AOSs has impeded the realization of fully complementary oxide circuits, low-power nonvolatile memory devices, and all-oxide neuromorphic hardware.

Recent studies have shown that tellurium 5p orbitals form delocalized valence bands with shallow acceptor states, which enhance hole transport even in amorphous tellurium-based oxide matrices [8]. This behavior contrasts with the highly localized oxygen 2p orbitals that typically limit hole mobility in oxide semiconductors. The extended nature of Te 5p orbitals supports more efficient hole conduction, challenging the usual view that AOSs suffer from poor charge transport due to disrupted pathways [9–11].

Tellurium-based materials have garnered significant attention as promising *p*-channel semiconductors due to their unique electronic structure, high intrinsic hole mobility, and favorable band alignment for hole injection. Among these, two-dimensional tellurium (2D-Te) stands out for its exceptional ambient stability and one of the highest hole mobilities reported in 2D materials, reaching values up to $\sim 600 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ [12, 13]. However, the narrow bandgap ($\sim 0.35 \text{ eV}$) of pristine 2D-Te severely limits its practical use, resulting in extremely low on/off current ratios and high leakage currents, which impede effective device switching and energy efficiency.

To overcome these fundamental limitations, controlled oxidation using ultraviolet–ozone (UV–O₃) treatment has emerged as an effective, low-temperature strategy. This process employs high-energy UV photons at 185 nm and 254 nm together with reactive oxygen species to induce uniform low-temperature, spatially uniform oxidation. Through this approach, 2D-Te is converted into amorphous tellurium trioxide (a-TeO₃), broadening the bandgap to $\geq 3 \text{ eV}$. This transformation stabilizes the electronic structure by achieving higher oxidation states and effectively suppresses leakage currents, enabling significantly improved on/off ratios and device performance [14].

AOSs exhibit remarkable sensitivity to oxygen stoichiometry and adsorbates. Their electrical properties, such as threshold voltage and hysteresis, vary significantly with oxygen content. Moreover, these properties can be strongly influenced by the measurement environment, such as vacuum versus ambient, due to changes in adsorbed species and defect dynamics [15, 16]. Persistent photocurrent (PPC) in AOS devices has been extensively investigated in *n*-channel materials such as a-IGZO, where it originates from photoinduced trap states, oxygen-vacancy (V_O) dynamics, and peroxide-related defects that give rise to long-lived carrier trapping and detrapping processes. These PPC mechanisms, which involve deep trap states below the conduction band and metastable donors, produce PPC decay on the minute-timescale and have been systematically characterized through temperature-dependent studies and wavelength-resolved optical measurements [17–19]. However, despite extensive research on environmental and photoinduced effects in AOSs, most studies have focused on *n*-type materials. In contrast, *p*-channel AOSs remain unexplored, leaving critical gaps in the understanding of charge-trapping physics and PPC mechanisms that are essential for the development of complementary memory and neuromorphic devices [11, 20, 21].

In this report, we demonstrate the high-performance *p*-channel AOS by converting crystalline 2D-Te into a-TeO₃ via a one-hour UV–O₃ treatment. The resulting device integrates transistor operation, nonvolatile memory, photodetection, and artificial synapse functionalities within a single platform. Structural analyses confirm complete amorphization with dominant Te⁶⁺ bonding and an increased bandgap (3.02 eV). The device exhibits a wide memory window ($> 58 \text{ V}$) with stable multilevel resistance states, driven by oxygen vacancy–adsorbate interactions. Under 390 nm illumination, it displays strong photoresponse, achieving a photoresponsivity of 411 A/W and a detectivity of 2.6×10^{11} Jones, alongside minute-timescale PPC and a pronounced light-induced contact potential difference (CPD) shift as measured by Kelvin probe force microscopy (KPFM). This PPC, consistent with the Madelung potential and peroxide models, enables robust synaptic behaviors such as long-term potentiation (LTP) and depression. Complementary synaptic transistors, implemented by integrating a-TeO₃ *p*-mode and monolayer molybdenum disulfide (1L-MoS₂) *n*-mode charge-trap memory (CTM) devices, establish a neuromorphic thermostat that mimics hypothalamic homeostasis. Ambient temperature fluctuations produce antagonistic synaptic responses under $\pm 15 \text{ V}$ gate pulses that autonomously activate cooling or heating pathways. In addition to thermoregulation, these complementary devices demonstrate light-programmable long-term memory and spike-dependent plasticity. When integrated into hardware-constrained neural networks, these complementary synaptic elements achieve MNIST classification accuracies comparable to ideal digital architectures, underscoring their potential as a scalable, low-power edge-AI platform.

2 | Results and Discussion

2.1 | Morphological, Spectroscopic, and Structural Evolution of 2D-Te During UV–O₃ Oxidation

Figure 1a shows the UV–O₃-induced transformation of crystalline 2D-Te. The pristine material (left) has a β -Te lattice with unit cell parameters of $0.45 \times 0.45 \times 0.59 \text{ nm}$. In this lattice, covalently bonded Te atoms assemble into pseudo-one-dimensional helical chains. These chains run parallel to the crystallographic *c*-axis, displaying a lattice spacing of 0.59 nm along the [001] direction [22, 23]. The 2D-Te nanosheets used in this study were synthesized via a hydrothermal process. The precise precursor composition, temperature-time profile, and post-growth treatments are provided in Figure S1. UV–O₃ oxidation was performed under simultaneous irradiation at 185 and 254 nm. Under these conditions, covalent Te–Te bonds gradually convert to Te–O–Te linkages, the long-range crystalline order collapses, and an amorphous tellurium-oxide network formed.

To investigate stoichiometric transformations and the evolution of the chemical state during UV–O₃-induced oxidation, we performed a comprehensive x-ray photoelectron spectroscopy (XPS) analysis comparing pristine crystalline 2D-Te with its oxidized form. Figure 1b shows the Te 3d XPS spectrum of 2D-Te revealing a complex multiphase chemical environment. The spectrum shows characteristic Te 3d_{5/2} and Te 3d_{3/2} doublet peaks separated by 10.4 eV, which accords with the spin-orbit coupling

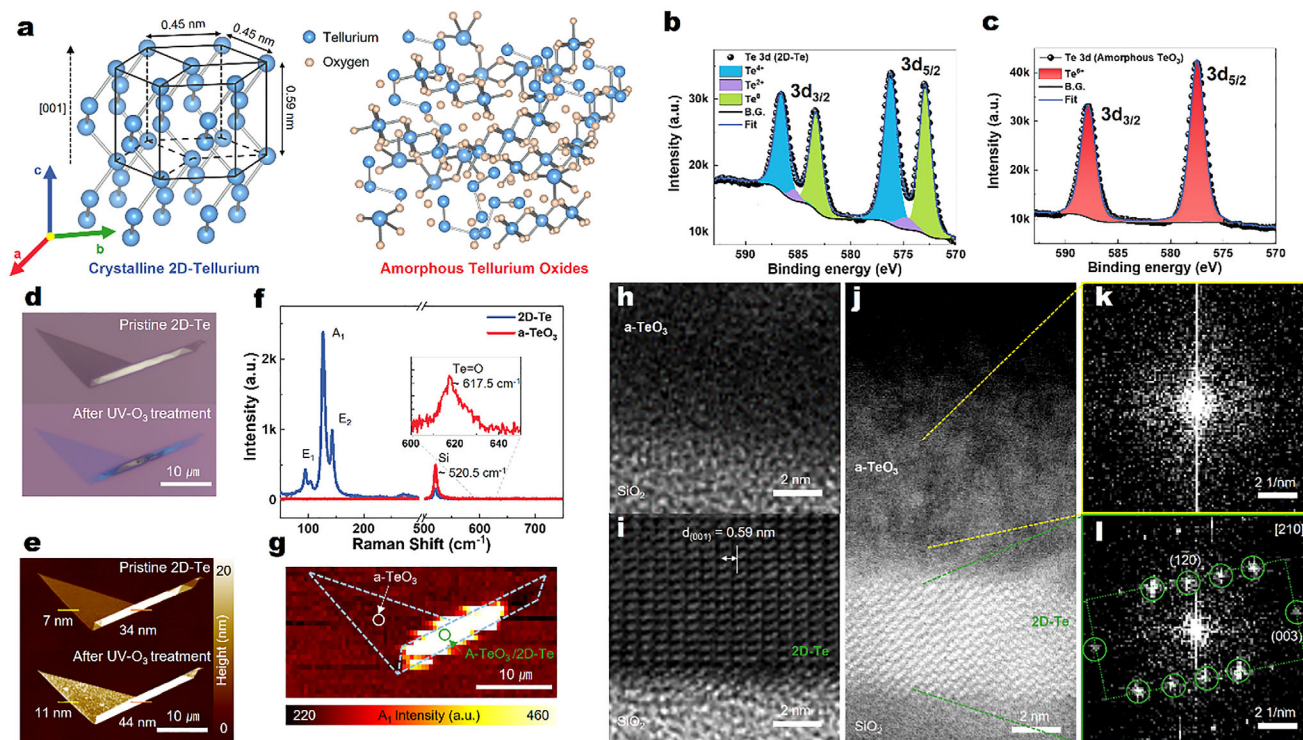


FIGURE 1 | (a) Crystal structure showing transformation from crystalline 2D-Te (left) with β -Te lattice to a-TeO₃ (right) after UV-O₃ treatment. (b) XPS Te 3d spectra of pristine 2D-Te reveal features assigned to multiple oxidation states (Te⁰, Te²⁺, Te⁴⁺). (c) After UV-O₃ exposure, the Te 3d XPS spectrum shows only Te⁶⁺, confirming complete oxidation to amorphous TeO₃. (d) Optical microscopy images showing enhanced transparency after 60 min UV-O₃ treatment. (e) AFM height measurements reveal thickness increase from 7–34 nm to 11–44 nm after oxidation. (f) Raman spectra show suppression of Te phonon modes (blue) and emergence of Te-O stretching (red) at 617.5 cm⁻¹ after treatment. (g) Raman intensity mapping of Te-O mode showing spatial distribution of oxide formation. HRTEM cross-sections: amorphous structure after treatment (h) versus crystalline lattice with 0.59 nm spacing in pristine 2D-Te (i). (j) Boundary between the surface a-TeO₃ layer and underlying crystalline 2D-Te. FFT analysis confirming amorphous structure (k) versus crystalline diffraction spots indexed to the (1-20) and (003) planes, viewed along the zone axis [210] of the β -Te crystal structure (l).

of tellurium. Detailed peak deconvolution reveals the coexistence of multiple oxidation states, including the dominant metallic Te⁰ component, which is centered at binding energies of ~572.9 (Te 3d_{5/2}) and 583.3 eV (Te 3d_{3/2}), and a Te⁴⁺ component at 576.2 and 586.6 eV (Te 3d_{5/2} and Te 3d_{3/2}, respectively) [24]. The presence of Te⁴⁺ species indicates that partial natural oxidation is occurring at the surface due to exposure to ambient air, forming native tellurium dioxide layers. Following 60 min of UV-O₃ treatment, the Te 3d spectrum undergoes dramatic simplification, as shown in Figure 1c. The complex, multiphasic spectrum transformed into a clean, monophasic profile that exclusively exhibited the Te⁶⁺ oxidation state. The binding energies shifted significantly to 577.5 eV (Te 3d_{5/2}) and 587.9 eV (Te 3d_{3/2}), which are characteristic of the highest oxidation state of tellurium found in tellurium trioxide. The complete absence of lower oxidation states (Te⁰, Te²⁺, and Te⁴⁺) confirmed the quantitative conversion of the surface layer to a-TeO₃ [14, 25]. This uniform distribution of oxidation states indicated that the UV-O₃ treatment achieved complete stoichiometric transformation, overcoming the kinetic limitations typically associated with conventional thermal oxidation processes. The transformation mechanism was further elucidated through O 1s spectral analysis, which revealed significant changes in oxygen bonding configurations accompanying the Te oxidation state evolution. In pristine 2D-Te, the limited oxygen content originated primarily from the native oxide layer with predominantly bridging oxygen configurations

characteristic of Te=O bonds in TeO₂. However, following UV-O₃ treatment, the O 1s spectrum shows a pronounced increase in non-bridging oxygen modes, as detailed in Figure S2 [26, 27].

Figure 1d shows optical microscopy images comparing 2D-Te (top) to the same sample after 60 min of UV-O₃ exposure (bottom). After treatment, thin 2D-Te flakes on a SiO₂ substrate exhibit markedly enhanced transparency, indicating an optical transformation of the material. Figure 1e illustrates the change in thickness quantitatively through atomic force microscopy (AFM) height measurements. Prior to treatment, the thinnest region measures ~7 nm, while the thickest region measures ~34 nm. After UV-O₃ oxidation, these values increased to ~11 and 44 nm, respectively. The increase in thickness demonstrated the expansion associated with the conversion of crystalline tellurium to amorphous tellurium oxide, reflecting the growth of the oxide layer. Figure 1f compares the Raman spectra of pristine 2D-Te and the same flake after 60 min of UV-O₃ treatment, both measured at room temperature. The 2D-Te exhibited four distinct phonon modes: the E₁ transverse optical phonon mode at 94 cm⁻¹, the E₁ longitudinal optical phonon mode at 106 cm⁻¹, the out-of-plane A₁ mode at 123 cm⁻¹, and the E₂ vibration at 143 cm⁻¹ (see Figure S3) [28].

The near-complete suppression of all Te-Te phonon modes during oxidation simultaneously indicated the loss of long-range

order and the weakening of the Te–Te bond network. Concurrently, a Raman peak emerges at 617.5 cm^{-1} , corresponding to the asymmetric stretching vibration of Te–O in a-TeO₃ [29]. After UV–O₃ treatment, the intensity of the Si substrate peak at 520.5 cm^{-1} in the Raman spectrum increased. This enhancement was attributed to the increased optical transparency of the underlying a-TeO₃ layer. The Raman signal from the underlying Si substrate was less attenuated and thus became more prominent as the crystalline 2D-Te was converted into a transparent amorphous oxide. Furthermore, ultraviolet photoelectron spectroscopy (UPS) confirmed the electronic density of states near the valence band edge, indicating bandgap widening (See Figure S4)

Figures 1h and 1i provide detailed structural insights into the phase transition of 2D-Te under UV–O₃ treatment (30 min), especially for samples less than 10 nm thick. Figure 1h shows a cross-sectional HRTEM image of a UV–O₃-treated sample that clearly revealed a fully a-TeO₃ layer on the surface, evidenced by the complete absence of periodic lattice planes. In contrast, Figure 1i shows a pristine 2D-Te sample with well-defined lattice planes and an interval of approximately 0.59 nm, which corresponds precisely to the (001) crystallographic plane of β -Te. This plane aligns with the helical chain direction and is a distinctive feature of high-quality, single-crystalline 2D-Te. Figure 1j further emphasizes the distinct boundary between the amorphous a-TeO₃ layer and the underlying crystalline 2D-Te with the thicker position (15 nm~). Figure 1k illustrates the fast Fourier transform (FFT) analysis of the amorphous region. The analysis reveals a diffuse halo with no distinct diffraction spots, confirming the complete loss of long-range order. Figure 1l, on the other hand, shows the FFT pattern from the crystalline 2D-Te region. This pattern displays clear, distinct diffraction spots that can be matched to the (120) and (003) planes of β -Te. These results demonstrate that UV–O₃ treatment amorphizes the surface of ultrathin 2D-Te without thermal heating [30].

2.2 | Environmental Effects on Hysteresis and Charge Trapping in Amorphous TeO₃ Transistors

Figures 2a and 2b depict the hysteresis behaviors of the same *p*-channel a-TeO₃ transistor under ambient and vacuum conditions, respectively. After deposition of 2D-Te, UV–O₃-induced amorphization was performed to form a uniform a-TeO₃ layer with a thickness of approximately 10 nm. OM images and detailed fabrication procedures are shown in Figure S5. The hysteresis window significantly expanded with increasing program/erase (P/E) back-gate voltage (V_{BG}) range from ± 10 to ± 50 V under ambient measurement conditions. Moreover, the maximum on current (I_{ON}) at the ± 50 V P/E condition decreased from $8.57\text{ }\mu\text{A}$ to $5.76\text{ }\mu\text{A}$, representing a 32.8% reduction. This decrease in I_{ON} further indicated the impact of charge trapping on carrier transport in the ambient environment. Figure 2c displays the field-effect mobility (μ_{FE}) and memory window as a function of P/E voltage under ambient and vacuum conditions, calculated by:

$$\mu_{FE} = \frac{L}{W C_{OX} V_{DS}} \times \frac{\partial I_{DS}}{\partial V_{BG}}, \quad (1)$$

where, $L = 6.31\text{ }\mu\text{m}$, $W = 5.89\text{ }\mu\text{m}$, $C_{OX} = 3.84 \times 10^{-8}\text{ F/cm}^2$, and $V_{DS} = 0.1\text{ V}$. The derivative $\partial I_{DS}/\partial V_{BG}$ is extracted from the transfer curves. From voltages above $\pm 20\text{ V}_{BG}$, stable memory operation was achieved through charge trapping. Compared with previously reported *p*-type oxide transistors, the a-TeO₃ device maintains competitive mobility (up to $72.6\text{ cm}^2\text{ V}^{-1}\text{ s}^{-1}$ in vacuum and $52.4\text{ cm}^2\text{ V}^{-1}\text{ s}^{-1}$ in ambient conditions at $\pm 30\text{ V}_{BG}$) while enabling stable memory operation under high gate voltages, which can be attributed to the relatively large bandgap of the oxide channel. In addition, relative to recently reported tellurium-oxide *p*-channel systems, our device uniquely combines competitive mobility with stable CTM operation in the high-voltage regime. (Supporting Information S6). Furthermore, Transfer curves measured under ambient air, vacuum, and dry N₂ conditions (Supporting Information S7) show that vacuum and N₂ responses are nearly identical, whereas the hysteresis is substantially enlarged in ambient air, supporting an adsorbate-assisted trapping mechanism. The memory window displayed in the bottom panel of Figure 2c, calculated as the absolute difference between forward and reverse gate voltages ($V_{forward} - |V_{reverse}|$) at a read current of $\sim 1\text{ nA}$, was substantially larger under ambient conditions compared to vacuum conditions. At the maximum P/E voltage of $\pm 50\text{ V}$, the memory window expanded to 58.5 V . The improved charge trapping resulting in a wider memory window is highly beneficial for the operation and stability of nonvolatile memory. Figure 2d shows the logarithmic output characteristics (I_{DS} vs V_{DS}) measured at V_{BG} ranging from $+50\text{ V}$ (off state) to -50 V (on state) in both ambient and vacuum conditions. The device exhibited a stable ohmic region, where the I_{DS} increased linearly with V_{DS} , in the range from $V_{DS} = 0$ to 1 V . In vacuum, the I_{ON} ($V_{BG} = -50\text{ V}$) was higher compared to ambient conditions.

To explain the origin of the enlarged memory window observed under ambient conditions due to increased charge trapping, the mechanisms are illustrated in Figures 2e and 2f. Figure 2e shows that V_0 in AOSs could be filled by atmospheric molecules, such as H₂O and O₂, which leads to a higher density of trapped charges [31]. To verify the intrinsic charge trapping phenomenon in AOS, Figure S8 demonstrates that even when the bottom dielectric was replaced with 2D hexagonal boron nitride, a distinct reduction in I_{ON} and an increase in hysteresis were clearly observed. When a negative gate voltage (①) is applied, sites that originally held electrons (thus carrying negative charge) become neutralized as holes are trapped. This charge neutralization by hole trapping leads to a sharp decrease in I_{DS} , resulting in a pronounced (wide) hysteresis in the transfer curves. Conversely, when a positive gate voltage (②) is applied, these traps are refilled with electrons. The repeated processes of charge trapping and detrapping during the forward sweep (①) and reverse sweep (②) give rise to the characteristic counterclockwise hysteresis observed in the transfer curve of a *p*-channel AOS transistor.

Figure 2g displays multilevel switching behavior of the device by sequentially applying erase and program pulses of varying amplitudes. The device could be set to four distinct low-resistance states (LRS1–LRS4) by applying erase pulses of increasing voltage ($+20, +30, +40, +50\text{ V}$), which gradually increased the I_{DS} . Conversely, sequential program pulses ($-20, -30, -40, -50\text{ V}$) reliably set the device to well-defined high-resistance states

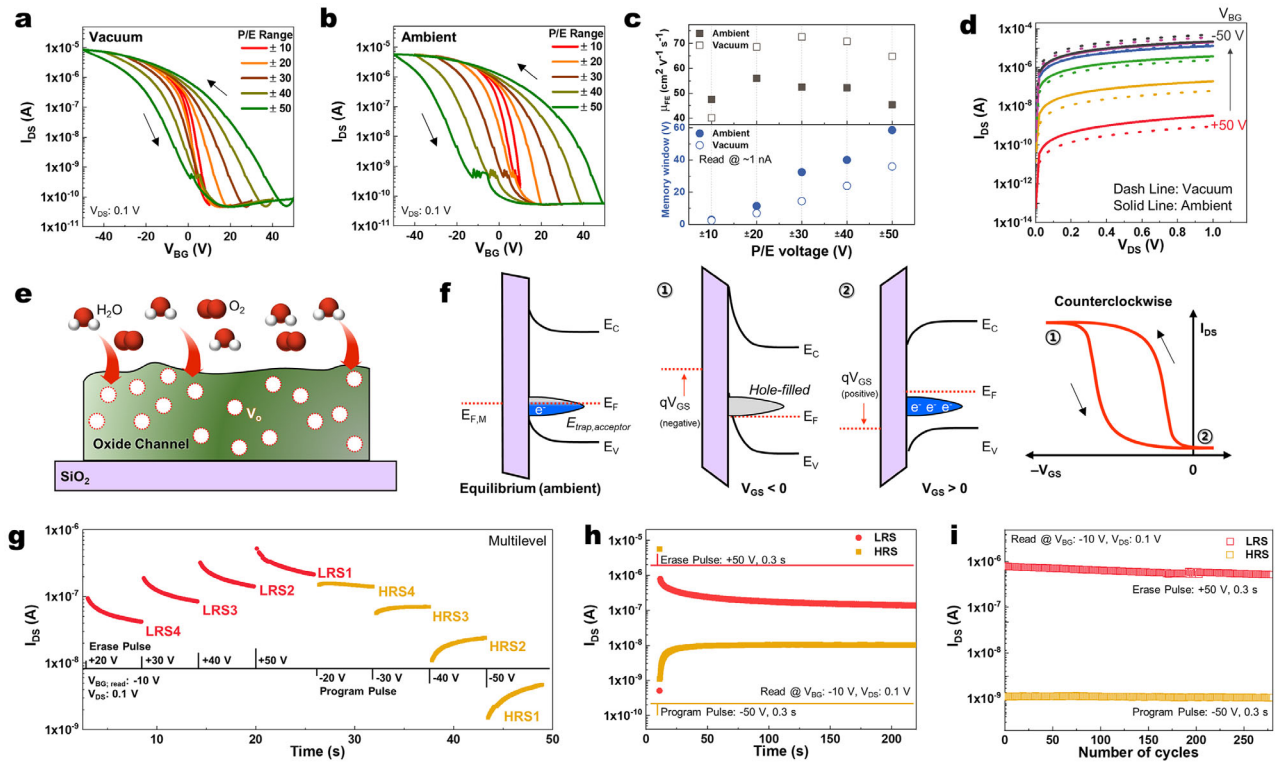


FIGURE 2 | Transfer characteristics of the *p*-channel a-TeO₃ transistor measured at $V_{DS} = 0.1$ V under ambient (a) and vacuum (b) conditions. Each panel shows five forward and reverse V_{BG} sweeps for increasing P/E voltage ranges of ± 10 , ± 20 , ± 30 , ± 40 , and ± 50 V. (c) Upper panel: μ_{FE} as a function of P/E voltage, with vacuum (open squares) and ambient (solid squares). Lower panel: Memory window calculated as ΔV at a read current of ~ 1 nA, with ambient (solid circles) and vacuum (open circles). (d) Logarithmic output characteristics measured at V_{BG} from +50 to -50 V in 20 V steps (Dashed lines: vacuum, solid lines: ambient). (e) Schematic illustration of adsorbates (H_2O , O_2) and V_O on the oxide channel. (f) Band diagrams (Ambient; Equilibrium, $V_{BG} < 0$ ①, $V_{BG} > 0$ ②) and transfer characteristic of the *p*-channel amorphous oxide semiconductor transistor. (g) Multilevel switching characteristics showing four distinct LRS1–LRS4 and HRS1–HRS4 achieved by applying different erase and program pulses. (h) Retention characteristics for LRS and HRS (i) Endurance performance over 250 P/E cycles, exhibiting stable LRS/HRS current levels.

(HRS1–HRS4), as evidenced by the corresponding decrease in I_{DS} . Figure 2h presents the data retention characteristics of both the LRS and HRS, monitored at a constant read bias ($V_{BG} = -10$ V, $V_{DS} = 0.1$ V, pulse time: 0.3 s) over an extended period. Following initialization by an appropriate erase or program pulse, both LRS and HRS exhibited stable I_{DS} values for over 200 s. Extended retention measurements (Figure S9) confirm that the programmed and erased states to $\sim 10^4$ s range with gradual current relaxation. Notably, a rapid change in current was observed during the initial 20 s, which was attributed to the fast release of charges trapped in shallow trap states. After this transient period, the current stabilized, indicating that deeper traps played the dominant role in long-term charge retention. Figure 2i shows the endurance characteristics of the device under more than 250 consecutive program/erase cycles, during which both LRS and HRS currents remained stable without any degradation. To demonstrate endurance of TeO₃ devices, HRS/LRS cycles was measured to 10^3 cycles in other TeO₃ transistors. The HRS and LRS remain clearly distinguishable and stable under cycled ± 50 V pulses, as shown in Figure S10. The environmental stability of the a-TeO₃ devices was further examined under controlled humidity conditions (Figure S11). the transfer curves show humidity-independence, as the on/off states and memory window remain clearly preserved, indicating stable charge-trap operation over a broad relative humidity range.

2.3 | CPD Evolution in a-TeO₃

To elucidate the photoinduced electronic properties, AFM and KPFM measurements of 2D-Te and a-TeO₃ were conducted in a dry nitrogen atmosphere. Figure 3a displays OM images capturing the transformation of a blue 2D-Te flake, initially deposited on a thermally evaporated Au film (inset), to a pale a-TeO₃ region after 60 min of UV exposure. AFM topography confirmed the structural transformation by revealing an increased thickness of 18.9 nm in the a-TeO₃ area, which exceeded the thickness 14–15 nm measured for pristine 2D-Te (Figure 3b). This expansion was consistent with oxygen incorporation and amorphization, reflecting successful nanoscale oxidation. Figure 3c presents the optical bandgap extraction of a-TeO₃ from confocal absorption spectroscopy using an indirect Tauc analysis. The measured absorbance spectrum was converted to an absorption coefficient, α , using the Beer–Lambert relation ($\alpha \propto A/t$), where t is the AFM-measured thickness (18.9 nm; Figure 3b). For disordered/amorphous oxides, the absorption edge is broadened by band-tail (Urbach-like) states and relaxed momentum selection, and the optical gap is commonly estimated using the Tauc relation for an allowed indirect transition, $(\alpha h\nu)^{1/2} = B(h\nu - E_g)$, where $h\nu$ is the photon energy and B is a proportionality constant. In our plot of $(\alpha h\nu)^{1/2}$ versus $h\nu$, two linear regimes are observed: a low-energy tail-absorption region (Linear Fit 1) and

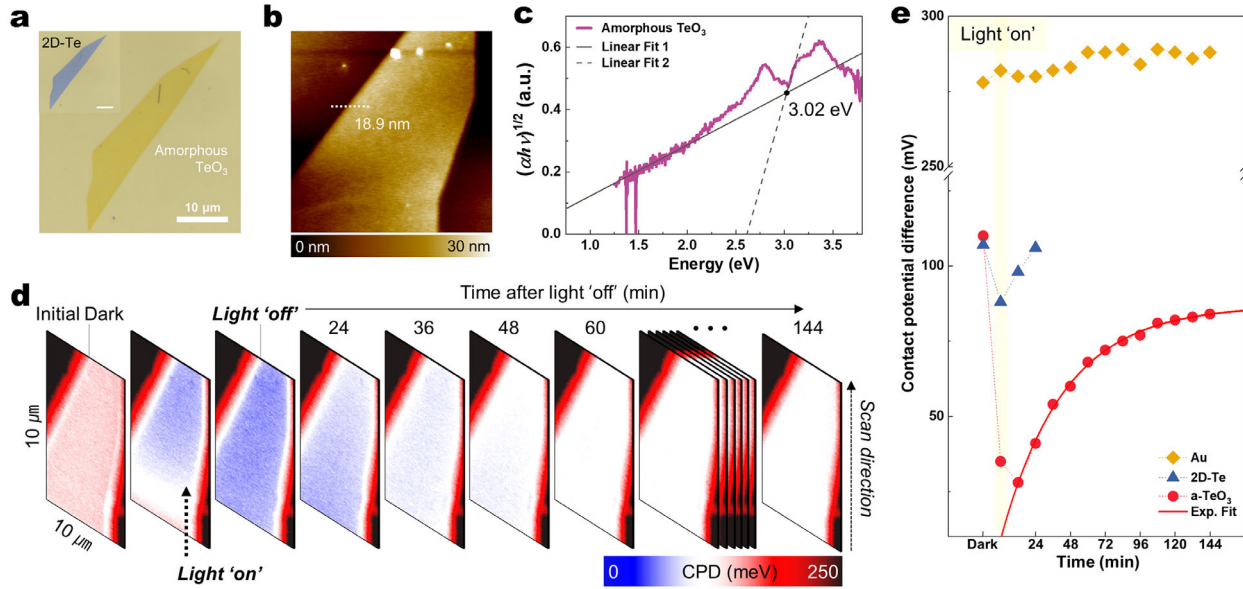


FIGURE 3 | (a) Optical microscope images showing the transformation from the initial blue 2D-Te flake (inset, on Au) to a pale a-TeO₃ region after 60 min of UV treatment; the scale bar is 10 μm. (b) AFM topography of a-TeO₃ displays a thickness of 18.9 nm, confirming structural expansion relative to 2D-Te. (c) The indirect Tauc plot for a-TeO₃, obtained by confocal absorption spectroscopy, reveals an optical bandgap of 3.02 eV. (d) Time-resolved KPFM mapping of a-TeO₃, where the initial state is obtained after more than 12 h in darkness. After light exposure, surface potential maps were acquired every 12 min over a total dark relaxation time of 144 min. The scan direction for KPFM is from bottom to top. (e) Evolution of CPD results, where Au (gold), 2D-Te (blue), and a-TeO₃ (red) traces are shown. After the light is switched off, CPD values were determined at 12-minute intervals up to 144 min; the data were fit with an exponential fitting line.

a higher-energy interband absorption region (Linear Fit 2). Tauc gap E_g is determined from the transition between these regimes by extrapolating the two linear fits and taking their intersection, yielding $E_g = 3.02$ eV for a-TeO₃.

Time-series KPFM mapping revealed light-induced CPD changes and its gradual recovery of a-TeO₃, as shown in Figure 3d. Initially, the measured CPD of a-TeO₃ was 110 mV in the dark state, which decreased to 35 mV under white LED illumination (power density: 10 mW/cm²), and partially recovered to 84 mV after 144 min in the dark state (Figure 3e). The slow CPD recovery resulted from persistent photoconductivity driven by carrier trapping in deep trap states. This phenomenon manifested as a pronounced optoelectronic memory effect: photogenerated carriers are captured in deep states, maintaining a nonequilibrium potential profile far beyond the illumination period. Exposure to white LED light induced a reversible CPD shift of 19 meV in crystalline 2D-Te. Once the light is turned off, the CPD value returns to its initial level within 24 min. Additionally, the Au film showed no measurable variation in the CPD upon light exposure. The KPFM measurement results indicate that the persistent quasi-Fermi level modulation observed in the a-TeO₃ originates from deep internal trap states, and that the gas adsorption/desorption effect is negligible.

2.4 | Optoelectronic Memory Operation Demonstrated by PPC-Based Light-Induced State (LIS) Switching

Figure 4a presents the photocurrent (I_{ph}) characteristics of the p-channel a-TeO₃ transistor as a function of V_{BG} sweep under

ambient UV illumination (wavelength 390 nm, power density 140 μW/mm²). The I_{ph} was determined by subtracting the dark current from the I_{DS} measured under illumination. As the P/E voltage sweep range was increased, both the magnitude of the photocurrent and the extent of hysteresis in the I_{ph} versus V_{BG} curves were significantly enhanced, indicating strong light-induced charge trapping effects in the device. To quantitatively evaluate the optoelectronic performance, the photoresponsivity (R) was calculated as:

$$R \text{ (A W}^{-1}\text{)} = \frac{I_{ph}}{P_{inc} \times A}, \quad (2)$$

where, R is the photoresponsivity, P_{inc} is the incident light power density, and A is the device area ($L = 6.31$ μm, $W = 5.89$ μm). Under a $V_{BG} = -30$ V, the maximum I_{ph} was 2.14 μA, yielding a photoresponsivity of 411 A/W. Photodetectivity (D) was determined by:

$$D \text{ (Jones)} = \frac{R\sqrt{A}}{\sqrt{2} \times q \times I_{Dark}}, \quad (3)$$

where, D is the detectivity, q is the elementary charge, and I_{Dark} is the dark current (here, 2.89 μA). The resulting detectivity was 2.6×10^{11} Jones. These results confirm that the device exhibits excellent gate-tunable optical memory and photodetection performance under ambient UV exposure.

Figure 4b shows the phenomenon of PPC in the p-channel a-TeO₃ transistor after exposure to UV light. Remarkably, when the UV illumination was turned off, the photocurrent did not immediately return to the original dark state. Instead,

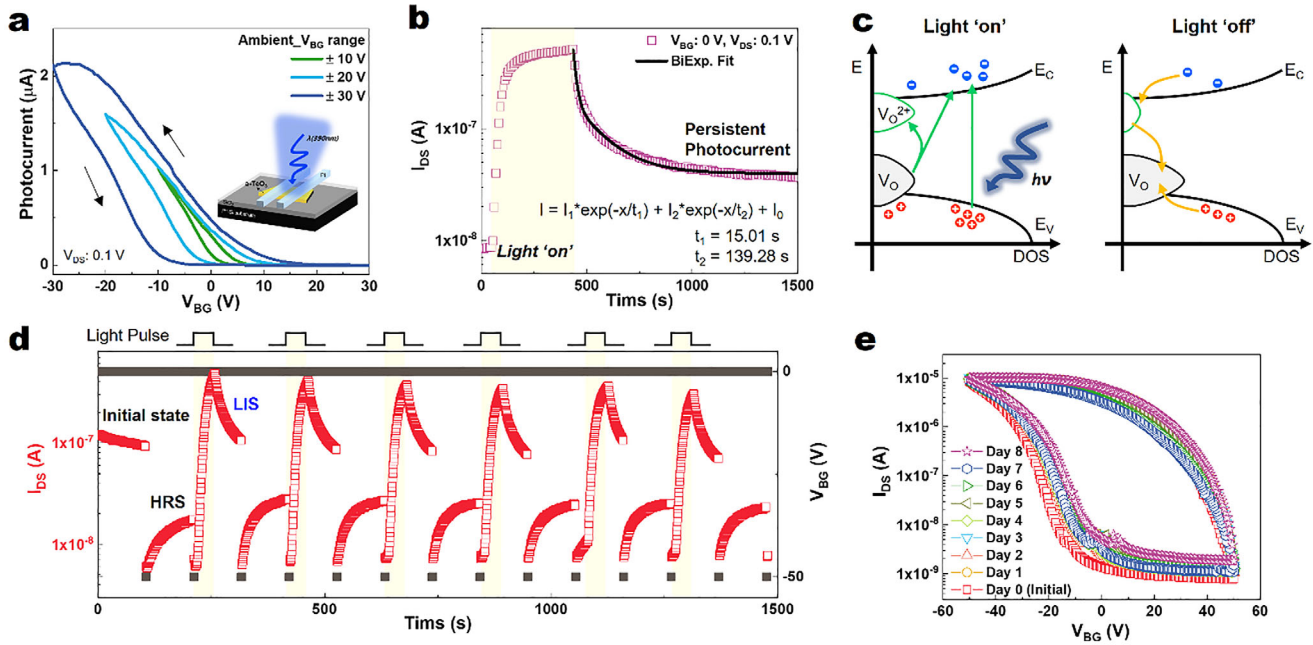


FIGURE 4 | (a) Transfer curves of the *p*-channel a-TeO₃ transistor ($V_{DS} = 0.1$ V) under ambient UV illumination (390 nm, 140 μ W/mm²) showing photocurrent. Sweeps were performed in a counterclockwise direction with the V_{BG} range expanded from ± 10 to ± 30 V. (b) Time-resolved I_{DS} measured at $V_{BG} = 0$ V, $V_{DS} = 0.1$ V. After ~ 50 s in the dark (light OFF), the device was illuminated for ~ 380 s (light on), then returned to the dark state (light OFF). (c) Schematic illustration of the PPC mechanism in *p*-channel amorphous oxide semiconductor. (d) Time-dependent photocurrent measurements confirming the HRS and LIS under ambient conditions. The device was measured at $V_{BG} = 0$ V and $V_{DS} = 0.1$ V with a 0.5 s read pulse. A 40 s light pulse was used to induce the LIS, and a program pulse ($V_{BG} = -50$ V) was applied for charge trapping. (e) Long-term stability assessment of *p*-channel a-TeO₃ transistor under ambient storage conditions. Transfer characteristics ($V_{DS} = 0.1$ V) measured systematically over 8 days while storing the device at atmospheric pressure, 20°C, and 40% relative humidity.

a significant fraction of the photogenerated current persisted for a prolonged period. The I_{DS} decay followed a biexponential function, $I = I_1 \exp(-x/t_1) + I_2 \exp(-x/t_2) + I_0$, indicating two distinct relaxation processes. The shorter time, constant $t_1 = 15.01$ s, corresponded to fast detrapping from shallow traps, while the longer time constant, $t_2 = 139.28$ s, was attributed to slow detrapping from deep traps. The pronounced long-lived current component highlighted the substantial retention of photocarriers, leading to the observed nonvolatile PPC effect.

Figure 4c illustrates a simplified PPC mechanism in *p*-channel AOS, based on the Madelung potential and peroxide model [32, 33]. Under UV illumination, neutral V_O and peroxide-related defects undergo photoionization to form doubly ionized oxygen defects (V_O^{2+}) which act as deep electron traps. When the light is turned off, no further free carriers are generated, and the electrons trapped in V_O^{2+} and other V_O -related states begin to be detrapped. Electrons in shallower traps are detrapped more rapidly, while electrons in deeper V_O^{2+} traps are emitted slowly, resulting in the long-lived PPC. This phenomenon is known as trap-assisted or Shockley-Read-Hall recombination [34]. The PPC could be erased by applying a sufficiently large negative V_{BG} , which induced strong band bending and injected electrons into the empty V_O^{2+} sites, neutralizing their trapping effect and suppressing the PPC immediately.

Figure 4d demonstrates the optoelectronic memory operation based on the PPC effect in *p*-channel AOS device. First, $V_{BG} = +50$ V ($V_{DS} = 0.1$ V) was applied to initialize the device to the HRS,

which remained stable after V_{BG} was returned to 0 V. A UV light pulse then generated a nonvolatile LIS through charge trapping in deep defects. Finally, a single negative gate pulse ($V_{BG} = -50$ V), reset the device to the original HRS by detrapping the stored charges. Figure 4e shows the remarkable long-term stability of the *p*-channel a-TeO₃ phototransistor under ambient conditions for 8 days. Stored at 20°C and 40% RH, the device maintained its switching ratio and modulation capability throughout, demonstrating robust operational reliability. Transfer measurements under 532 nm illumination (Figure S12) show nearly no change from the dark state, indicating that sub-gap excitation at 532 nm does not effectively induce charge trapping in a-TeO₃.

2.5 | Bioinspired Synaptic Architecture and Autonomous Optical LTP

Figure 5a depicts the biological archetype of synaptic transmission: a presynaptic neuron communicates with a postsynaptic partner across a synaptic cleft. In our artificial synapse, electrical and/or optical spikes served the role of neurotransmitter release, modulating the device conductance (synaptic weight) through photogating and charge-trapping/detrapping physics. Optoelectronic synapses routinely exploited PPC and trap dynamics to emulate short- and long-term plasticity, which formed the conceptual basis of the measurements. Figure 5b demonstrates that LTP can be achieved using optical stimulation alone (UV light at 140 μ W/mm²), without requiring electrical assistance. This finding was significant as it revealed the high photosensitivity

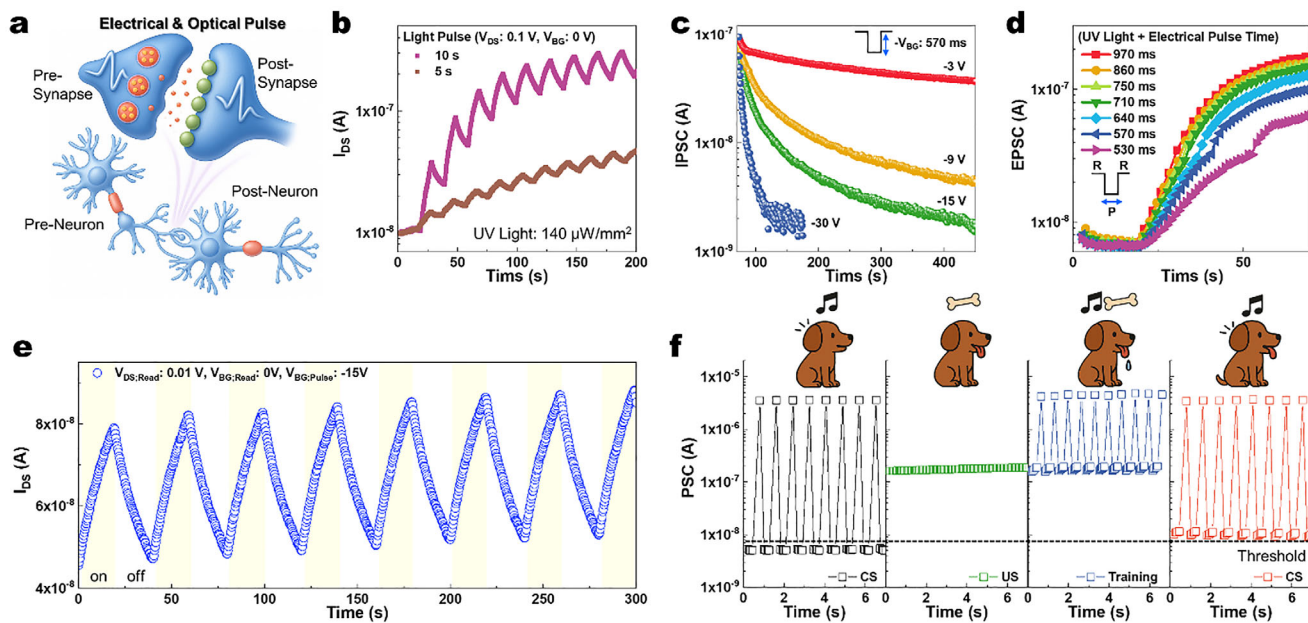


FIGURE 5 | (a) Schematic illustration of the artificial synaptic device mimicking pre- and post-synaptic neuron connections with electrical and optical pulse stimulation. (b) LTP achieved through optical stimulation alone under UV light exposure (140 $\mu\text{W}/\text{mm}^2$) over 200 s. Measurements performed at $V_{DS} = 0.1$ V and $V_{BG} = 0$ V with light on time = 10 s (purple) and 5 s (brown). (c) IPSC behavior with systematic V_{BG} variation from -3, -9, -15, -30 V using 570 ms pulses, demonstrating progressive inhibition enhancement over 400 s readout at $V_{DS} = 0.1$ V, $V_{BG} = 0$ V. (d) EPSC enhancement with increasing V_{BG} pulse duration (530 to 970 ms) at $V_{BG} = -15$ V, demonstrating pulse width-dependent potentiation. Readout performed at $V_{DS} = 0.1$ V, $V_{BG} = 0$ V. (e) Long-term stability under repetitive cycling ($V_{DS;\text{Read}} = 0.01$ V, $V_{BG;\text{Read}} = 0$ V, $V_{BG;\text{Pulse}} = -15$ V) maintaining consistent postsynaptic current over 300 s of continuous operation. Light on/off cycles of 20 s each demonstrate stable switching behavior without performance degradation. (f) Hardware implementation of Pavlovian conditioning showing four phases with readout performed at $V_{DS} = 0.1$ V, $V_{BG} = 0$ V: CS baseline with V_{BG} pulse voltage = -9 V ($<10^{-8}$ A), US response under UV light (140 $\mu\text{W}/\text{mm}^2$), training phase with simultaneous CS-US pairing (V_{BG} pulse = -9 V + UV light), and enhanced CS response (V_{BG} pulse = -9 V only) above threshold after associative learning.

and autonomous optical programming capability of the device. The I_{DS} increased systematically with light exposure duration, reaching values from $\sim 10^{-8}$ to $\sim 10^{-7}$ A, confirming successful potentiation through PPC accumulation.

Figure 5c demonstrates inhibitory postsynaptic current (IPSC) behavior using different negative V_{BG} pulses with 570 ms pulse duration, with readout performed at $V_{BG} = 0$ V. The variation of V_{BG} (-3, -9, -15, and -30 V) showed progressive enhancement of inhibitory behavior, as evidenced by the decreasing IPSC amplitude over time within the ~ 400 s measurement window. This gate-controlled IPSC precisely mimics biological inhibitory synapses, which are crucial for implementing balanced excitatory-inhibitory networks essential for neural computation. More negative V_{BG} values created stronger depletion fields that reduce hole concentration in the p -type channel, leading to enhanced inhibitory responses. The gate-tunable inhibition strength provided precise control over synaptic depression, complementing the LTP functionality and enabling bidirectional synaptic plasticity. The ability to achieve multiple levels of inhibition through gate voltage modulation showed that the device could perform multilevel synaptic weighting, which was a critical requirement for implementing complex neural networks.

Figure 5d reveals the critical relationship between electrical pulse duration and excitatory postsynaptic current (EPSC) amplitude enhancement, where longer pulse durations from 530 to 970 ms progressively increase synaptic responses. To support the synaptic

functionality beyond pulse-duration-dependent EPSC modulation, additional interval/intensity-dependent paired-pulse facilitation (PPF) and pulse-number-dependent EPSC measurements were performed under ambient conditions, as shown in Figures S13 and S14. The PPF ratio compared with the pulse interval indicates well-defined postsynaptic response. Further, larger pulse intensity shows PPF index increment, connected to charge trapping behavior. Repeated-pulse measurements also showed progressive EPSC accumulation with increasing pulse number, confirming repeated-stimulus-induced synaptic weight modulation in the a- TeO_3 device. This behavior resulted from the distinctive characteristics of PPC in trap-rich oxide systems, where extended illumination enabled progressive trap saturation and corresponding increases in decay time constants. Unlike conventional frequency-dependent plasticity where higher frequencies improve responses, this device demonstrated pulse duration-dependent enhancement through trap filling effects, effectively emulating the functional outcome of high-frequency stimulation.

Figure 5e demonstrates hardware implementation of Pavlovian associative learning through four distinct experimental phases. The initial conditioned stimulus (CS) alone phase established a baseline response of approximately 10^{-8} A, while the unconditioned stimulus (US) alone phase confirmed strong pathway functionality. During the training phase, repeated CS-US pairing established temporal association through asymmetric spike-timing-dependent plasticity mechanisms. The critical

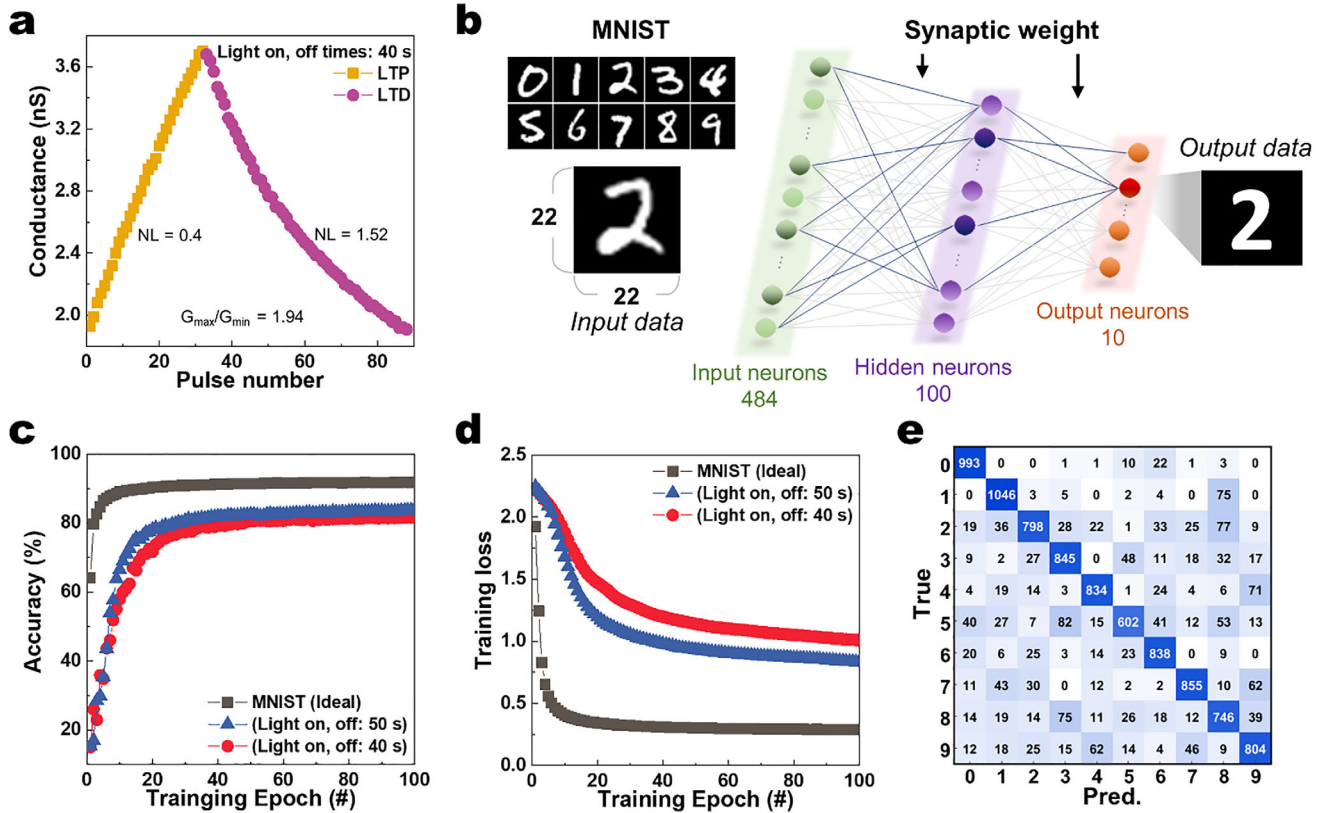


FIGURE 6 | (a) Conduction modulation of the a-TeO₃-based charge trapping synaptic transistor, measured with light on/off times of 40 s. The device was read with $V_{BG,Read} = 0$ V and $V_{DS} = 0.01$ V, $V_{BG,Pulse} = -15$ V, 570 ms pulses were applied. LTP and LTD behavior are shown with extracted NL factors of 0.4 and 1.52, and a conductance ratio of $G_{max}/G_{min} = 1.94$. (b) Schematic of the three-layer neural network for MNIST digit classification using 484 memristive synapses, 100 hidden neurons, and 10 output neurons. (c) Training accuracy over epochs for each implementation. Networks were trained with experimental LTP/LTD quantized updates reflecting device-based pulse timing (on/off: 50 s and 40 s). Early epochs show rapid accuracy increase for all cases. (d) Training loss curves corresponding to Figure (c), showing smooth loss decay and optimization for both device-based and ideal network implementations. (e) Confusion matrix for quantized memristor network.

observation occurs in the final CS post-training phase, where enhanced responses above the original threshold indicated successful associative memory formation. Figure 5f shows long-term operational stability under repetitive cycling conditions with parameters $V_{DS,Read} = 0.01$ V, $V_{BG,Read} = 0$ V, and $V_{BG,Pulse} = -15$ V. The consistent postsynaptic current amplitude of approximately 8×10^{-8} A maintained over 300 s of continuous on-off switching demonstrated excellent endurance and reliability.

2.6 | Implementation of a-TeO₃-Based Charge Trapping Synaptic Transistor for MNIST Digit Recognition

Figure 6 presents a comprehensive evaluation of charge trapping transistor-based synaptic plasticity and its implementation in a three-layer perceptron neural network for MNIST digit recognition. Figure 6a shows how the a-TeO₃-based charge trapping transistor enables synaptic plasticity by combining electrical and optical stimulation. Device conductance was measured under a $V_{BG,Read} = 0$ V and a $V_{DS} = 0.01$ V, with periodic gate pulses ($V_{BG,Pulse} = -15$ V, duration: 570 ms). During the measurement sequence, the device was illuminated with 390 nm light for 40 s, inducing PPC and increasing its conductance. The light-

driven conductance enhancement served as an analog of LTP in artificial synapses because photogenerated carriers became trapped at defect sites or within the charge-trapping layer, thereby strengthening the synaptic weight. After turning off the light for another 40 s and the electrical pulses continued, the device conductance gradually decreased, corresponding to LTD. The linearity and nonlinearity (NL) of these weight updates were essential metrics for hardware-based neural networks. Following the analytical model established by Xu Liping et al. the NL of conductance changes during LTP and LTD was quantified by fitting the experimental curves to the following equations: [35].

$$\text{for LTP } G_{n+1} = G_n + \Delta G_p = G_n + \alpha_p \exp \left(-\beta_p \frac{G_n - G_{min}}{G_{max} - G_{min}} \right), \quad (4)$$

$$\text{for LTD } G_{n+1} = G_n + \Delta G_d = G_n - \alpha_d \exp \left(-\beta_d \frac{G_{max} - G_n}{G_{max} - G_{min}} \right), \quad (5)$$

where, G_n and G_{n+1} represent the conductance values measured immediately after the n^{th} and $(n+1)^{\text{th}}$ stimulation pulses, respectively. The parameters G_{min} and G_{max} corresponded to

the measured minimum and maximum conductance of the synaptic device, defining the dynamic range of synaptic weight modulation. The coefficient α described the nominal step size, meaning the initial change in conductance produced by each stimulation pulse before nonlinear effects became significant. The NL parameters, β_p (for LTP) and β_d (for LTD), were determined by fitting the experimentally measured conductance trajectories to the analytical model. The NL factors extracted from the fitting are $\beta_p = 0.4$ for potentiation and $\beta_d = 1.52$ for depression. The device achieved a dynamic conductance ratio ($G_{\max}/G_{\min}$) of 1.94, demonstrating its ability to represent a broad spectrum of synaptic weights.

Figure 6b depicts an artificial neural network architecture employing a-TeO₃-based charge trapping synaptic transistors for hardware-embedded neuromorphic learning. In this network, each pixel of a 22×22 MNIST image was mapped to one of 484 input neurons, which were connected to 100 hidden neurons and finally to 10 output neurons representing the digit classes 0–9. Each synaptic connection was realized by an a-TeO₃ transistor, and its conductance was updated in discrete steps to emulate biological synaptic behavior. During network training, the synaptic weights are limited to the measured conductance levels, reflecting finite programming resolution and the nonideal switching dynamics. The evolution of synaptic conductance after P pulses, whether for LTP or LTD, followed device-specific exponential update rules: [36, 37].

$$\begin{aligned} G_{\text{LTP}} &= G_{\min} + \frac{(G_{\max} - G_{\min})(1 - e^{-P/\text{NL}})}{1 - e^{-P_{\max}/\text{NL}}}, \\ G_{\text{LTD}} &= G_{\max} - \frac{(G_{\max} - G_{\min})(1 - e^{-(P - P_{\max})/\text{NL}})}{1 - e^{-P_{\max}/\text{NL}}}, \end{aligned} \quad (6)$$

where, $P_{\max}$ is the total pulse count for the full dynamic range. Figures 6c and 6d show a comparison of an MNIST classifier built with an a-TeO₃-based charge-trapping synaptic transistor against an ideal digital reference and two distinct, device-realistic implementations. The synaptic weights were updated according to pulse-dependent, experimentally determined LTP/LTD rules and quantized into discrete conductance steps, reflecting the limited programming resolution. Condition A (blue), with longer optical gating on/off times (50 s), achieved higher programming granularity than Condition B (red; 40 s) because the residence time under illumination directly modulated the conductance increment of optoelectronic memristor. This behavior has been consistently reported in recent neuromorphic optoelectronic studies [38].

Throughout the training process, accuracy increases rapidly during the initial epochs while the loss decreases smoothly. Both hardware-inspired networks closely follow the trajectory of the ideal digital model. These results align with previous findings that soft-bounded exponential update schemes and noise-/asymmetry-aware learning mechanisms effectively compensate for variability at the device level [39]. Recent studies have also demonstrated tellurium oxide-based artificial optoelectronic synapses with broadband responsiveness, further supporting the potential of Te-oxide platforms for neuromorphic and optoelectronic applications [40].

Figure 6e shows the confusion matrix that summarized the digit classification performance of the a-TeO₃-based charge-trapping synaptic transistor network for the MNIST task. The confusion matrix exhibited a diagonal structure, suggesting that each output neuron responded selectively to its designated digit class. Neurons with partially overlapping receptive fields exhibited correlated selectivity, which produced ambiguity for similar stimuli. The charge-trapped memristive network reproduced key aspects of real neural processing, as shown by the alignment between our synaptic-weight distributions, the resulting neuron activations, and the observed confusion structure.

2.7 | Neuromorphic Autonomous Based on Complementary *p*- and *n*-Mode CTM Devices

Figure 7 shows a neuromorphic autonomous system inspired by the brain, which uses light-induced *p*-channel CTM technology to emulate the homeostatic principles by which the human brain automatically maintains body temperature at around 37°C (98.6 °F). The hypothalamus orchestrates human thermoregulation by functioning as an autonomous that continuously compares the current body temperature with the physiological set point of 37°C [41]. Figure 7a presents temperature fluctuation patterns that reflect environmental temperature variations experienced in daily life. The red circle indicates a high-temperature region (26°C), which activates warm-sensitive neurons in the hypothalamus. These neurons initiate cooling responses through sweating and vasodilation. The blue circle indicates a low-temperature region (20°C). This region triggers cold-sensitive neurons, which promote heating responses via muscle contractions (shivering) and brown adipose tissue thermogenesis. The neuronal architecture on the right mimics the cellular temperature autonomous mechanism, in which brain cells receive temperature information through dendrites and control heating and cooling pathways through synaptic connections.

Figure 7b presents the electronic circuit implementation of the biological neural network, where presynaptic pulses (V_{in}) independently modulate 1L-MoS₂-based *n*-type CTM and a-TeO₃-based *p*-type CTM devices to operate heating and cooling systems. Figure 7c illustrates that both device types share a common gate electrode, enabling complementary responses to identical input stimuli and forming antagonistic synaptic pairs that emulate biological synaptic behavior. Figures 7d and 7e illustrate the complementary synaptic dynamics of *p*-mode and *n*-mode charge-trap devices under light-induced PPC. In this homeostatic control system (Figures 7d and 7e), +15 V_{BG} pulses were generated when the current temperature exceeded the set threshold (requiring cooling), while −15 V_{BG} pulses were triggered when the temperature falls below the threshold (requiring heating). When the high temperature was sensed and cooling was required, the *p*-type a-TeO₃ CTM device operated as a “cooler” and responds to +15 V V_{BG} pulses by detrapping holes. This results in a prominent EPSC that actively drove cooling (Figure 7d). Conversely, in response to −15 V_{BG} pulses, holes became trapped, which suppressed channel conduction and resulted in an IPSC. Under cold conditions requiring heating, the *n*-mode 1L-MoS₂ CTM device functions as a ‘heater’ (Figure 7e). Applying a −15 V_{BG} pulse to the *n*-mode CTM facilitated electron detrapping and triggered a pronounced EPSC, which activated the heating system

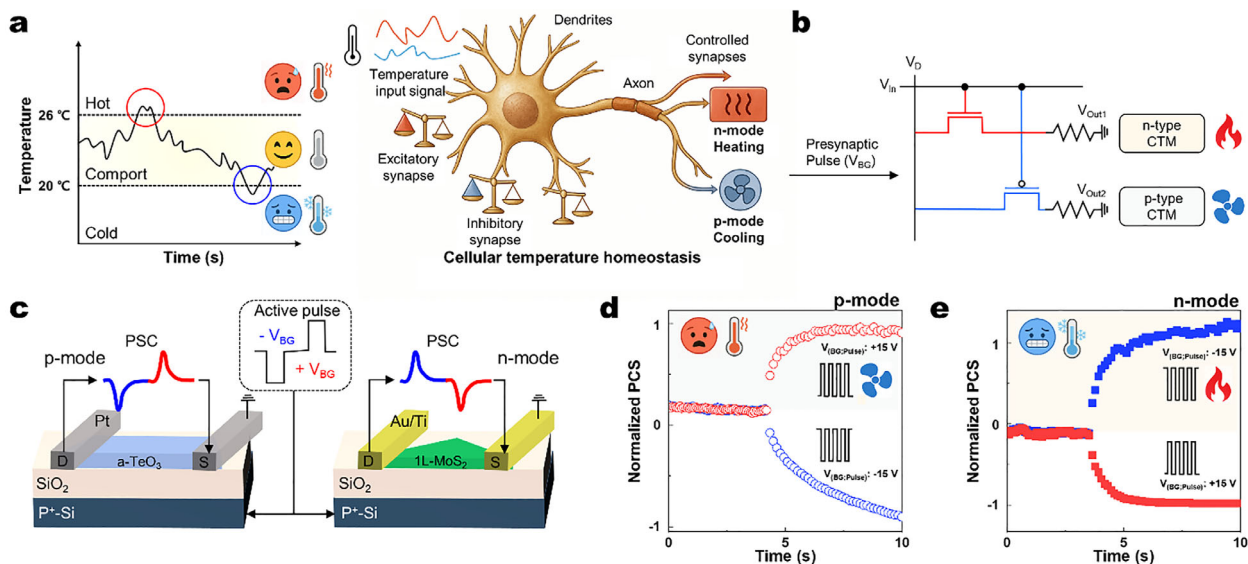


FIGURE 7 | (a) Environmental temperature fluctuations (20°C–26°C) trigger opposing physiological responses: warm-sensitive pathways activate cooling, cold-sensitive pathways activate heating, emulating hypothalamic autonomous. (b) Circuit implementation uses a common presynaptic V_{BG} pulse to drive a-TeO₃ p -mode CTM for cooling and a 1L-MoS₂ n -mode CTM for heating. (c) Both devices share a gate electrode and exhibit complementary PSC modulation under $\pm V_{BG}$ pulses. (d) Under PPC state with $V_{DS} = 0.1$ V and $V_{BG, read} = 0$ V, a -15 V pulse causes the p -mode device to trap holes, exhibiting IPSC (blue circle); at $+15$ V, hole detrapping yields EPSC (red circle). (e) Conversely, the n -mode device shows EPSC at -15 V (blue square) and IPSC at $+15$ V (red square).

and maintained autonomous. Switching the pulse polarity to $+15$ V_{BG} re-trapped the electrons, resulting in an IPSC and reduced conduction. Figure S15 provides a detailed characterization of the 1L-MoS₂ n -mode CTM and its PPC. This complementary charge-trap mechanism mimicked the antagonistic neural pathways that mediate biological thermoregulation, enabling autonomous bidirectional temperature regulation with precise autonomous. Moreover, both devices supported light-induced long-term memory transitions and spike-dependent synaptic plasticity, which facilitated energy-efficient adaptive learning and homeostatic control.

3 | Conclusion

This study presents a comprehensive neuromorphic platform, spanning materials synthesis to system-level demonstration, by converting crystalline 2D-Te into fully amorphous p -type oxide semiconductor TeO₃ through UV-O₃ treatment, and employing it as the active medium in charge-trapping synaptic transistors. Comprehensive characterization via high-resolution XPS, Raman spectroscopy, AFM/HRTEM, and FFT analysis confirms amorphization with dominant Te⁶⁺ bonding, highlighting the speed and uniformity advantages of UV-O₃ over conventional thermal oxidation. According to environmentally dependent electrical measurements, vacancy-mediated charge trapping under ambient condition significantly broadens the hysteresis and memory windows. Furthermore, when photon energy exceeding the band gap is applied, the device behavior exhibits trap-assisted transport and PPC, consistent with the Madelung-potential and peroxide models that describe high-performance p -type AOSs. A complementary neuromorphic thermostat is realized through the combined operation of p -mode and n -mode CTM devices, faithfully reproducing hypothalamic autonomous by switching

cooling and heating pathways through IPSC/EPSC responses. Together with light-programmable long-term memory and spike-dependent plasticity, these results provide clear design principles for next-generation light-programmable CTM devices.

4 | Experimental Section/Methods

4.1 | Phase Transformation of 2D-Te to a-TeO₃: Oxidation Mechanism

The controlled oxidation mechanism operates through a synergistic dual-wavelength photochemical process in a UV-ozone treatment system (Ossila) equipped with a mercury lamp emitting at 185 (646 kJ mol⁻¹) and 254 nm (472 kJ mol⁻¹). At 185 nm, atmospheric oxygen is photolyzed into highly reactive atomic oxygen ($O_2 + h\nu \rightarrow 2O$), which then undergoes three-body collisions to form ozone ($O + O_2 + M \rightarrow O_3 + M$), where M is a third-body collision partner such as N₂ or O₂ that dissipates excess energy to stabilize the O₃ molecule. Concurrently, 254 nm radiation decomposes ozone ($O_3 + h\nu \rightarrow O_2 + O^*$) and directly photoactivates oxidation reactions on the Te surface, establishing a sustained reactive O^{*}/O₃ flux that continuously supplies oxidizing species. Because the photon energies at both wavelengths exceed the electron affinity of tellurium (190.02 kJ mol⁻¹), they efficiently promote electron excitation and subsequent oxidation pathways, progressively converting 2D-Te to a-TeO₃, with ozone-derived atomic oxygen further accelerating the transformation.

4.2 | Fabrication and Characterization of a-TeO₃ Transistors

The a-TeO₃ transistors were fabricated on 90 nm SiO₂ gate dielectric layers using high WF platinum (Pt, 40 nm)

electrodes patterned by photolithography and sputter deposition, and their electrical characteristics were measured with a Keithley 4200 system under both ambient and low-vacuum conditions (~ 5 mTorr).

KPFM Measurements: Surface topography and CPD mapping were conducted using an atomic force microscope (NX10, Park Systems) integrated within a nitrogen-purged glove box to eliminate ambient environmental effects. Both 2D-Te and a-TeO₃ specimens were prepared on Au-coated substrates to ensure stable electrical grounding. Measurements employed NSG01/Pt-coated cantilevers (NT-MDT) with a scan area of $10 \times 10 \mu\text{m}^2$ (256×256 pixels) at a scan rate of 0.35 Hz. The CPD calibration was performed using a highly ordered pyrolytic graphite (HOPG, SPI Supplies) standard.

4.3 | XPS Measurements

XPS measurements were carried out using a Theta Probe spectrometer (Thermo Fisher Scientific) equipped with a monochromatic Al K α radiation source ($h\nu = 1486.7$ eV, energy resolution: 0.85 eV) under ultra-high vacuum ($\sim 10^{-9}$ Torr). High-resolution spectra were collected at a 90° take-off angle using a pass energy of 100 eV and a dwell time of 50 ms per step. Peak fitting and deconvolution were performed with Thermo Avantage software following smart background subtraction.

4.4 | Confocal Raman Spectroscopy

Raman spectra were acquired using a Horiba LabRAM HR Evolution confocal Raman microscope equipped with a 532 nm linearly polarized laser. The excitation beam was focused onto the sample through a $100\times$ Olympus objective lens (NA = 0.9), with the laser power limited to 220 μW to avoid thermal or photo-induced damage. Scattered light was dispersed by a spectrometer with an 1800 g/mm grating and detected using a charge-coupled device, providing a spectral resolution of 0.6 cm^{-1} . To ensure optimal signal quality and measurement reproducibility, data were collected with progressively increased acquisition times.

4.5 | Network architecture and training protocol

The MNIST dataset used for training (60 000 images) and testing (10 000 images) was center-cropped from the original 28×28 pixel images to 22×22 pixels to match the input layer dimensions. This preprocessing step reduces noise from background pixels at the edges of the images and improves computational efficiency by decreasing the input size from 784 to 484 [42]. All pixel values were normalized without further preprocessing to ensure compatibility with memristor conductance mapping. The network was implemented using NeuroPack, with the memristor device model inheriting from the BaseMemristor interface, where LTP and LTD rules were explicitly defined. ReLU activation was applied in the positive linear layers to enforce nonnegative weights, ensuring synaptic connections remained within the physically realizable conductance range. Training was conducted using stochastic gradient descent with a learning rate of 0.01

over 100 epochs and a batch size of 128, balancing computational efficiency and gradient stability.

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Conflicts of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supporting File: sml174713-sup-0001-SuppMat.docx.